

MADISON COOTS

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RESEARCH FOCUS

Responsible AI, inequality, discrimination, and the use of computational methods to advance decision-making and policy across a variety of domains, including in healthcare, lending, criminal justice, and education.

EDUCATION

Harvard Kennedy School <i>Ph.D. in Public Policy</i>	Cambridge, MA August 2022 - Present
Harvard University <i>M.A. in Public Policy</i>	Cambridge, MA August 2022 - May 2025
Stanford University <i>M.S. in Computer Science</i> <i>Specialization: Artificial Intelligence</i>	Stanford, CA September 2019 - June 2021
Stanford University <i>Bachelor of Science - Management Science and Engineering</i> <i>Minor: English</i>	Stanford, CA September 2015 - June 2019

PEER-REVIEWED PUBLICATIONS

- [1] **Automated Court Date Reminders Reduce Warrants for Arrest: Evidence from a Text Messaging Experiment.** Alex Chohlas-Wood, **Madison Coots**, Joe Nudell, Julian Nyarko, Emma Brunskill, Todd Rogers, and Sharad Goel. *Science Advances*. 2025.
- [2] **Racial Bias in Clinical and Population Health Algorithms: A Critical Review of Current Debates.** **Madison Coots**, Kristin A. Linn, Sharad Goel, Amol S. Navathe, and Ravi B. Parikh. *Annual Review of Public Health*. 2025.
- [3] **A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk.** **Madison Coots**, Soroush Saghaian, David Kent, and Sharad Goel. *Annals of Internal Medicine*. 2024.
- [4] **Learning to be Fair: A Consequentialist Approach to Equitable Decision-Making.** Alex Chohlas-Wood, **Madison Coots**, Henry Zhu, Sharad Goel, and Emma Brunskill. *Management Science*. 2024.
- [5] **Designing Equitable Algorithms.** Alex Chohlas-Wood*, **Madison Coots***, Julian Nyarko*, and Sharad Goel*. *Nature Computational Science*, Vol. 3. 2023. *Authors contributed equally.

BOOK CHAPTERS

- [1] **Racial Bias in Clinical and Population Health Algorithms: A Critical Review of Current Debates.** **Madison Coots**, Kristin A. Linn, Sharad Goel, Amol S. Navathe, and Ravi B. Parikh. Chapter in *Social Stratification, 5th Edition* (2027). Edited by Michelle Jackson, Nima Dahir, Claire Daviss, and David B Grusky. Routledge.

RESEARCH IN PROGRESS

- [1] **Algorithmic Bias in Lending: Evidence from a Fintech Audit.** **Madison Coots**, Robert Bartlett, Julian Nyarko, and Sharad Goel. *Working paper*. 2026.
- [2] **Incorporating Text in Empirical Audits of Police Searches.** **Madison Coots**, Hans Gaebler, Josh Grossman, Alex Chohlas-Wood, Julian Nyarko, and Sharad Goel.
- [3] **AI Tutoring and Teacher Support Tools for Elementary Education**, with Calvin Isley, Elisa Xi Chen, Nick Masi, Joe Nudell, Charlotte Tuminelli, Teddy Svoronos, and Sharad Goel.

HONORS, AWARDS, AND FELLOWSHIPS

Dissertation Research Grant, \$10,000	2026
• Horowitz Foundation for Social Policy	
Rising Star in IOE-ISyE-MS&E	2026
• Workshop jointly hosted by University of Michigan, Georgia Tech, and Stanford University	
Dean's Award for Excellence in Student Teaching	2025
• Harvard Kennedy School of Government	
Distinction in Student Teaching Award	2025
• Harvard Kennedy School of Government	
Social Equity and Health Equity Grant, \$5,000	2025
• Malcolm Weiner Center for Social Policy, Harvard Kennedy School of Government	
James M. and Cathleen D. Stone PhD Scholar in Inequality and Wealth Concentration	2023 - Present
• Stone Program in Wealth Distribution, Inequality, and Social Policy, Harvard University	
Harvard Graduate Prize Fellowship	2022 - 2023
• Harvard University	
Stanford Engineering Coterminial Fellowship	2019 - 2020
• Stanford University	
Stokes Scholar	2017 - 2021
• U.S. Federal Government	

SELECTED PRESENTATIONS

Econometric Society ESIF: Economics and AI+ML Conference	Ithaca, NY
<i>"Algorithmic Bias in Lending: Evidence from a Fintech Audit"</i> (Submitted talk)	2026
Production and Operations Management Society Annual Conference	Reno, NV
<i>"A Profit-Based Measure of Lending Discrimination"</i> (Submitted talk)	2026
IOE-ISyE-MS&E Rising Stars Workshop	Ann Arbor, MI
<i>"A Profit-Based Measure of Lending Discrimination"</i> (Invited talk and poster)	2026
American Law and Economics Association Conference	New York, NY
<i>"A Profit-Based Measure of Lending Discrimination"</i> (Submitted talk)	2025
Production and Operations Management Society Annual Conference	Atlanta, GA
<i>"A Profit-Based Measure of Lending Discrimination"</i> (Invited talk)	2025
Predictive Analytics & Comparative Effectiveness Center Symposium	Boston, MA
<i>"A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk"</i> (Invited talk)	2024
Society of Medical Decision Making 46th Annual Meeting	Boston, MA
<i>"A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk"</i> (Submitted talk)	2024
International Conference on Computational Social Science	Philadelphia, PA
<i>"A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk"</i> (Poster)	2024
Computational and Methodological Statistics Conference	Remote
<i>"A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk"</i> (Invited talk)	2023
APPAM Fall Research Conference	Atlanta, GA
<i>"Automated Court Date Reminders Reduce Warrants for Arrest"</i> (Submitted talk)	2023
INFORMS General Meeting	Phoenix, AZ
<i>"A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk"</i> (Invited talk)	2023
INFORMS Healthcare Conference	Toronto, ON
<i>"A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk"</i> (Invited talk)	2023
ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization	Arlington, VA
<i>"Learning to be Fair"</i> (Submitted talk with Alex Chohlas-Wood)	2022

TEACHING

Harvard Kennedy School of Government

- **Teaching Fellow.** API 201: Quantitative Analysis and Empirical Methods I (Fall 2024, 2025). *Graduate course in applied statistics: exploring and summarizing data with R, probability theory, statistical inference, decision analysis.*
- **Teaching Fellow.** API 202Z: Quantitative Analysis and Empirical Methods II (Spring 2025, 2026). *Advanced graduate course in applied statistics: statistical inference and linear regression models.*
- **Teaching Fellow.** API 203Z: Quantitative Analysis and Empirical Methods III (Spring 2025). *Advanced graduate course in applied statistics: causal inference and machine learning.*
- **Teaching Fellow.** DPI 681M: The Science and Implications of Generative AI (Spring 2024) *Designed and taught the technical complement to the course on language models, deep learning models, and transformers.*

Stanford University, School of Engineering

- **Course Assistant.** MS&E 125: Applied Statistics (Winter 2020) *Undergraduate course in applied statistics: exploring and summarizing data, statistical inference, linear and logistic regression models.*
- **Course Assistant.** MS&E 252: Foundations of Decision Analysis (Fall 2019). *Graduate course in quantitative decision analysis covering: utility theory, sensitivity analysis, value of information. Recognized by Stanford Center for Professional Development for excellence in teaching.*

Stanford Code in Place Program

- **Section Leader.** (Spring 2020). *Part of a teaching team for Code in Place, offered by Stanford during COVID-19 pandemic, with 10,000 global students and 900 volunteer teachers participating from around the world. Prepared and taught a weekly discussion section of 10-12 students to supplement professors' lectures in a 5-week introductory online Python programming course.*

PROFESSIONAL SERVICE

Ad Hoc Peer Reviewer

- *Epidemiology* (2025)
- *International Conference on Computational Social Science* (2025, 2026)

Organizing Team Member

- *Teaching Statistics in the Age of Agentic AI*, Harvard Kennedy School (June 2026)

Supporting Instructor

- *An Introduction to Reproducible Programming and Project Management*, Society of Medical Decision Making Annual Meeting (2024)

SKILLS SUMMARY

Languages: R, Python

AI: LLM prompt and response evaluation, text data processing and information extraction, chatbot design and user experience for AI systems

Methods: Stochastic modeling, decision analysis, statistical inference, causal inference

PROFESSIONAL EXPERIENCE

Senior Data Scientist (Part-time)

Systems & Technology Research

Woburn, MA

Feb. 2023 - Present

Data Scientist

Stanford Computational Policy Lab

Boston, MA

Sep. 2020 - Aug. 2022

Data Scientist (Part-time)

Aerospace Technical Services

Remote

Sep. 2020 - Jun. 2024

Data Science Intern

U.S. Federal Government (agency withheld)

Washington D.C.

Jun. 2017 - Jan. 2021